

Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	1.000000000e+03	kg m ⁻³
Dynamic viscosity	ν	1.000000000e+01	Pa s
Kinematic viscosity	ν	1.000000000e-02	m ² s ⁻¹
Characteristic velocity	$u_{\max}$	1.000000000e+00	m s ⁻¹
Characteristic length	D	1.000000000e+00	m
Reynolds number	Re	1.000000000e+02	dimensionless
Body force	G	9.810000000e+00	m s ⁻²
Grid points in x	N_x	3.100000000e+01	cells
Grid points in y	N_y	3.100000000e+01	cells
Grid spacing in x	Δx	3.22580645e-02	m
Grid spacing in y	Δy	3.22580645e-02	m
Time step	Δt	1.000000000e-04	s
Numerical threshold	ϵ	7.000000000e-11	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
Parameters listed above are the values used by the selected solver.			

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	[(5435, 80, 80), (6613, 80, 80)]
Regression field shape (t, x, y)	[(3435, 76, 76), (4613, 76, 76)]
Regression samples	528,645,530,945,280
Dictionary terms	5
First saved time index	150

Quantity	Value
Temporal crop at each boundary	1000 layers
x-boundary crop	2 points
y-boundary crop	2 points

4. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

Learned equation

$$u_t = -1.10882883e+00*uu_x - 9.00486192e-01*vu_y + 1.17132003e-02*u_{xx} + 9.54792967e-03*u_{yy} - 9.61052126e-04*p_x$$

Term	Reference coefficient	Learned coefficient	Percentage error
uu_x	-1.00000000e+00	-1.10882883e+00	-1.08828833e+01%
vu_y	-1.00000000e+00	-9.00486192e-01	-9.95138077e+00%
u_xx	+1.00000000e-02	+1.17132003e-02	1.71320035e+01%
u_yy	+1.00000000e-02	+9.54792967e-03	4.52070326e+00%
p_x	-1.00000000e-03	-9.61052126e-04	-3.89478738e+00%

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-08
Selected tolerance	9.42668455e-04
L0 penalty	1.00000000e-06
Training RMSE	5.54856252e-02
Validation RMSE	2.76317905e-02

6. Reproducibility Notes

- Input directory:
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output/4movingwalls_80,
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output/lidcavity_81
- Fields were loaded starting at saved time index 150.

- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.

7. Transverse Momentum Equation

Reference equation

$$v_t = -1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e-02*v_{xx} + 1.00000000e-02*v_{yy} - 1.00000000e-03*p_y$$

Learned equation

$$v_t = -1.03471651e+00*uv_x - 1.11093148e+00*vv_y + 9.53136692e-03*v_{xx} + 1.38013045e-02*v_{yy} - 1.05880897e-03*p_y$$

Term	Reference coefficient	Learned coefficient	Percentage error
uv_x	-1.00000000e+00	-1.03471651e+00	-3.47165138e+00%
vv_y	-1.00000000e+00	-1.11093148e+00	-1.10931478e+01%
v_xx	+1.00000000e-02	+9.53136692e-03	4.68633079e+00%
v_yy	+1.00000000e-02	+1.38013045e-02	3.80130446e+01%
p_y	-1.00000000e-03	-1.05880897e-03	-5.88089681e+00%

V STRidge Selection and Error Metrics

Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	4.52023912e-02
Validation RMSE	1.64496948e-02